

Permutation and Combination

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Permutation

- An arrangement or listing in which order or placement is important is called a permutation .

Simple example:

“combination lock”

$31 - 5 - 17$

is NOT the same as

$17 - 31 - 5$

Though the same numbers are used, the order in which they are turned to, would mean the difference in the lock opening or not. Thus, the order is very important.

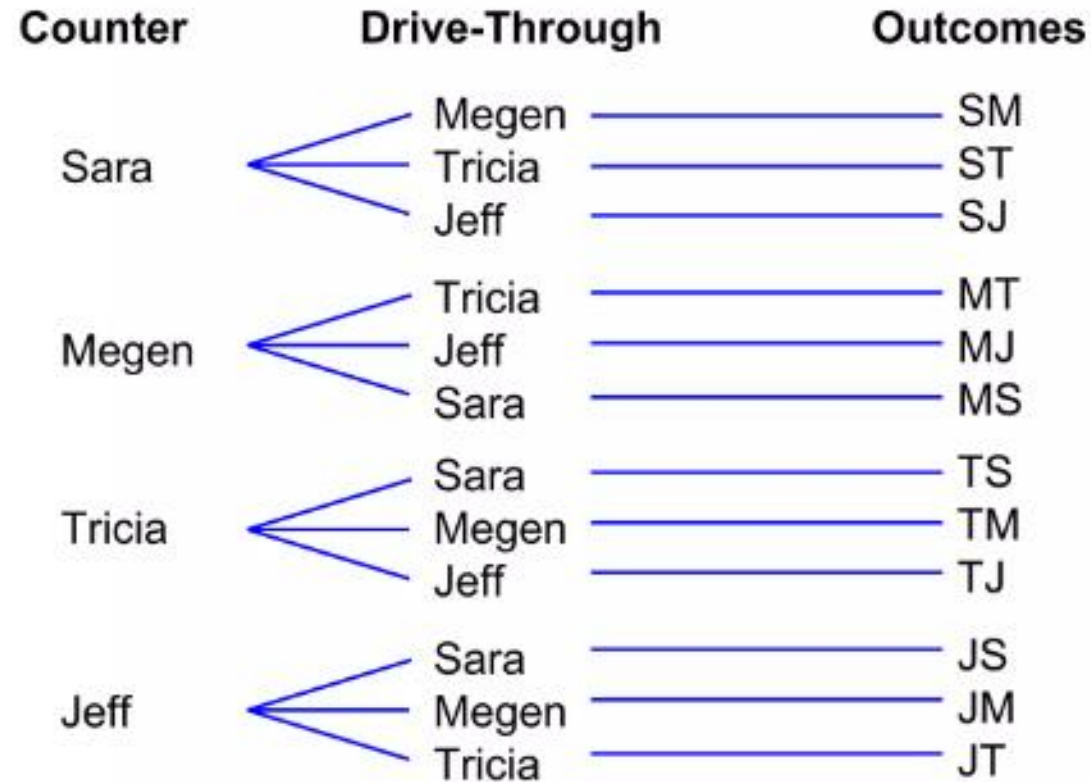


Combination
lock?

Example

- The manager of a coffee shop needs to hire two employees, one to work at the counter and one to work at the drive-through window. Sara, Megen, Tricia and Jeff all applied for a job. How many possible ways are there for the manager to place the applicants?

Solution



Solution

- In the previous example, the positions are in specific order, so each arrangement is unique.
- The symbol $4 P 2$ denotes the number of permutations when arranging 4 applicants in two positions.

SM ST SJ MT MJ MS TS TM TJ JS JM JT Outcomes

Example

- Find the permutation of $10P6$?

151200

Permutation and probability

- A computer program requires the user to enter a 7-digit registration code made up of the digits 1, 2, 4, 5, 6, 7, and 9. Each number has to be used, and no number can be used more than once.
1. How many different registration codes are possible?
 2. What is the probability that the first three digits of the code are even numbers?

7P7

Example

What is the probability that the first three digits of the code are even numbers?

First choice	3	$= 3 * 2 * 1 * 4 * 3 * 2 * 1$ $= 144$
Second choice	2	
Third choice	1	
Fourth choice	4	$P(3 \text{ digits}) = \frac{P(E)}{P(S)}$
Fifth choice	3	$= \frac{144}{5040}$
Six choice	2	$= 3\%$
Seven choice	1	

Combination

- An arrangement or listing in which order is not important is called a combination .
- For example
if you are choosing 2 salad ingredients from a list of 10, the order in which you choose the ingredients does not matter.

$10C2$

$$\frac{10!}{(10 - 2)! 2!} = 45$$

Example

- The students of Mr. Fant's Seminar class had to choose 4 out of the 7 people who were nominated to serve on the Student Council. How many different groups of students could be selected?
- Note

Solution

$7C_4$

$$\frac{7!}{(7-4)!4!}$$

$$=35$$

Difference between permutation and Combination

- The order in which the people are being chosen does not matter because the positions for which they are being chosen are the same. They are all going to be members of the student council, with the same duties. (Combination)
- if Mr. Fant's class was choosing 4 out of 7 students to be president, vice-president, secretary, and treasurer of the student council, then the order in which they are chosen would matter. (Permutation)

Example

- A group consists of 4 girls and 7 boys. In how many ways can a team of 5 members be selected if the team has
 - (i) no girls
 - (ii) at least one boy and one girl
 - (iii) at least three girls

Solution

- Number of girls = 4
- Number of boys = 7

1. No girls

$$\begin{aligned}\text{Total number of ways the team can have no girls} &= {}^4C_0 \times {}^7C_5 \\ &= 1 \times 21 \\ &= 21\end{aligned}$$

Solution (Cont...)

(ii) at least one boy and one girl

$$1 \text{ boy and 4 girls} = {}^7C_1 \times {}^4C_4 = 7 \times 1 = 7$$

$$2 \text{ boys and 3 girls} = {}^7C_2 \times {}^4C_3 = 21 \times 4 = 84$$

$$3 \text{ boys and 2 girls} = {}^7C_3 \times {}^4C_2 = 35 \times 6 = 210$$

$$4 \text{ boys and 1 girl} = {}^7C_4 \times {}^4C_1 = 35 \times 4 = 140$$

$$\begin{aligned} \text{Total number of ways the team can have at least one boy and one girl} &= \\ &7 + 84 + 210 + 140 \end{aligned}$$

$$= 441$$

Solution (Cont...)

(iii) At least three girls

Total number of ways the team can have at least three girls

$$= {}^4C_3 \times {}^7C_2 + {}^4C_4 \times {}^7C_1$$

$$= 4 \times 21 + 7$$

$$= 84 + 7$$

$$= 91$$